



JABchem



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Past Papers Nat 5 Chemistry

2017 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/100)	%	
A	74+	74%	35.2%
B	63+	63%	21.8%
C	52+	52%	19.4%
D	46+	46%	8.6%
No award	<46	<46%	15.0%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	14.6 /20	35.5 /60	14.7 /20

2017 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning								
1	D	91	$\text{Rate} = \frac{\Delta \text{quantity}}{\Delta \text{time}} = \frac{2\text{g}}{30\text{s}} = 0.0667 \text{ g s}^{-1}$								
2	C	90	☒ A if atomic number = 24 then number of protons = 24 ☒ B if atomic number = 45 then number of protons = 45 ☒ C atomic number = no of protons = 21 and mass no. = protons + neutrons = 21 + 24 = 45 ☒ D if atomic number = 24 then number of protons = 24								
3	A	67	Dichromate ion formula is $\text{Cr}_2\text{O}_7^{2-}$ $\therefore$ $\text{Cr}_2\text{O}_7^{2-}$ ion must be balanced by a 2+ ion $\therefore$ Zn^{2+}								
4	D	60	☒ A Magnesium melts at 650°C $\therefore$ temperature is 730°C so magnesium is liquid ☒ B Magnesium melts at 650°C $\therefore$ temperature is 730°C so magnesium is liquid ☒ C Magnesium has density 1.74 g cm^{-3} and floats in magnesium chloride (2.32 g cm^{-3}) ☒ D Magnesium formed is a liquid at 730°C and floats in molten magnesium chloride as magnesium metal has a lower density.								
5	A	58	A base is a chemical which neutralises an acid to form water e.g. metal hydroxides (alkalis), metal oxides and metal carbonates are bases								
6	B	72	Spectator ions appear chemically unchanged on both sides of a chemical equation: $\text{Ag}^+_{(\text{aq})} + \text{NO}_3^-_{(\text{aq})} + \text{K}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})} \rightarrow \text{AgCl}_{(\text{s})} + \text{K}^+_{(\text{aq})} + \text{NO}_3^-_{(\text{aq})}$ $\xrightarrow{\text{K}^+ \text{ appears on both sides of equation}}$ $\xleftarrow{\text{NO}_3^- \text{ appears on both sides of equation}}$								
7	C	86	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$								
8	B	70	$\text{gfm} = \frac{\text{mass}}{\text{no. of mol}} = \frac{7\text{g}}{0.25\text{mol}} = 28\text{g mol}^{-1}$ ☒ A gfm $\text{C}_2\text{H}_6 = (2 \times 12) + (6 \times 1) = 24 + 6 = 30\text{g}$ ☒ B gfm $\text{C}_2\text{H}_4 = (2 \times 12) + (4 \times 1) = 24 + 4 = 28\text{g}$ ☒ C gfm $\text{C}_3\text{H}_8 = (3 \times 12) + (8 \times 1) = 36 + 8 = 44\text{g}$ ☒ D gfm $\text{C}_3\text{H}_6 = (3 \times 12) + (6 \times 1) = 36 + 6 = 42\text{g}$								
9	A	56	☒ A no of mol = volume $\times$ concentration = $0.1 \text{ litres} \times 0.4 \text{ mol l}^{-1} = 0.04 \text{ mol}$ ☒ B no of mol = volume $\times$ concentration = $0.2 \text{ litres} \times 0.3 \text{ mol l}^{-1} = 0.06 \text{ mol}$ ☒ C no of mol = volume $\times$ concentration = $0.3 \text{ litres} \times 1.0 \text{ mol l}^{-1} = 0.3 \text{ mol}$ ☒ D no of mol = volume $\times$ concentration = $0.4 \text{ litres} \times 0.5 \text{ mol l}^{-1} = 0.2 \text{ mol}$								
10	A	85	☒ A C_7H_{16} gives a general formula of $\text{C}_n\text{H}_{2n+2}$. Alkanes have general formula $\text{C}_n\text{H}_{2n+2}$. ☒ B C_7H_{14} gives a general formula of C_nH_{2n} . Alkanes have general formula $\text{C}_n\text{H}_{2n+2}$. ☒ C C_7H_{12} gives a general formula of $\text{C}_n\text{H}_{2n-2}$. Alkanes have general formula $\text{C}_n\text{H}_{2n+2}$. ☒ D C_7H_{10} gives a general formula of $\text{C}_n\text{H}_{2n-4}$. Alkanes have general formula $\text{C}_n\text{H}_{2n+2}$.								
11	D	83	☒ A X is an alkene $\therefore$ has $\text{C}=\text{C}$ double bond $\therefore$ will decolourise bromine solution quickly ☒ B Y is a cycloalkane $\therefore$ no $\text{C}=\text{C}$ double bond $\therefore$ does not decolourise bromine solution ☒ C Y is a cycloalkane $\therefore$ no $\text{C}=\text{C}$ double bond $\therefore$ does not decolourise bromine solution ☒ D X (alkene) will decolourise bromine and Y (cycloalkane) will not decolourise								
12	C	50	<table border="1"> <thead> <tr> <th>Element in Fuel</th><th>Carbon</th><th>Hydrogen</th><th>Sulphur</th></tr> </thead> <tbody> <tr> <td>Product of Combustion</td><td>Carbon Dioxide</td><td>Water</td><td>Sulphur Dioxide</td></tr> </tbody> </table>	Element in Fuel	Carbon	Hydrogen	Sulphur	Product of Combustion	Carbon Dioxide	Water	Sulphur Dioxide
Element in Fuel	Carbon	Hydrogen	Sulphur								
Product of Combustion	Carbon Dioxide	Water	Sulphur Dioxide								
13	C	81	Vinegar is a dilute solution of ethanoic acid CH_3COOH								
14	B	92	☒ A endothermic reactions absorb energy from the surroundings (drops the temp) ☒ B exothermic reactions release energy to the surroundings (raises the temp) ☒ C energy required to start reaction not related to the energy given off ☒ D a reaction must have an energy change to be exothermic								

15	D	77	☒ A Structure shown is a covalent network ☒ B Structure shown is an ionic lattice ☒ C Structure shown is a molecular covalent ☒ D Structure shown is a metallic lattice
16	C	79	☒ A tin is a medium reactivity metal and is found combined in the Earth's crust ☒ B magnesium is a upper reactivity metal and found combined in the Earth's crust ☒ C gold is a very unreactive metal and found uncombined in the Earth's crust ☒ D sodium is a very reactivity metal and found combined in the Earth's crust
17	A	70	☒ A Oxygen is not an essential element for healthy plant growth ☒ B Nitrogen in an essential element for plants and found in fertilisers ☒ C Potassium in an essential element for plants and found in fertilisers ☒ D Phosphorus in an essential element for plants and found in fertilisers
18	B	95	☒ A Nitric acid is made by the Ostwald Process ☒ B Ammonium is made by the Haber Process ☒ C Alkenes can be made by the dehydration reaction of alcohols by removing water ☒ D Esters are made by the condensation of alcohols and carboxylic acids
19	A	64	☒ A Barium sulphate is an insoluble salt and can be made by a precipitation reaction ☒ B Lithium nitrate is soluble and cannot be made by a precipitation reaction ☒ C calcium chloride is soluble and cannot be made by a precipitation reaction ☒ D ammonium phosphorus is soluble and cannot be made by a precipitation reaction
20	C	40	A standard solution is a solution whose concentration is accurately known and used in titration to work out the number of moles or concentration of a second substance.

2017 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning																
1a	Isotopes	Same atomic number but different mass number number of protons number of neutrons																
1b	Different number of neutrons in each	<table><tr><th>Isotope</th><th>Protons</th><th>Neutrons</th><th>Electrons</th></tr><tr><td>³⁶₁₈ Ar</td><td>18</td><td>36-18 = 18</td><td>18</td></tr><tr><td>³⁸₁₈ Ar</td><td>18</td><td>38-18 = 20</td><td>18</td></tr><tr><td>⁴⁰₁₈ Ar</td><td>18</td><td>40-18 = 22</td><td>18</td></tr></table>	Isotope	Protons	Neutrons	Electrons	³⁶ ₁₈ Ar	18	36-18 = 18	18	³⁸ ₁₈ Ar	18	38-18 = 20	18	⁴⁰ ₁₈ Ar	18	40-18 = 22	18
Isotope	Protons	Neutrons	Electrons															
³⁶ ₁₈ Ar	18	36-18 = 18	18															
³⁸ ₁₈ Ar	18	38-18 = 20	18															
⁴⁰ ₁₈ Ar	18	40-18 = 22	18															
1c	36	Relative atomic mass is the average mass of the different isotopes of the same element in a sample. As the average mass is 36.2 of isotopes with masses of 36, 38 and 40 the mass abundant isotope must be 36 as the average is closest to 36.																
2a	Nanotube	Problem Solving: Selecting information from passage																
2b	Lithium	Problem Solving: Selecting information from passage																
2c	20.5	m=41g gfm H ₂ = 2g mol ⁻¹ no of moles = $\frac{\text{mass}}{\text{gfm}}$ = $\frac{41\text{g}}{2\text{g mol}^{-1}}$ = 20.5 mol																
3a	Diagram showing:																	
3b	tetrahedral	<table><tr><td>HCl</td><td>H₂O</td><td>NH₃</td><td>CH₄</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td>Linear</td><td>Angular</td><td>Trigonal Pyramidal</td><td>Tetrahedral</td></tr></table>	HCl	H ₂ O	NH ₃	CH ₄					Linear	Angular	Trigonal Pyramidal	Tetrahedral				
HCl	H ₂ O	NH ₃	CH ₄															
Linear	Angular	Trigonal Pyramidal	Tetrahedral															
3c	Gains one electron	$\begin{array}{ccc} \text{Cl} & + e^- & \longrightarrow & \text{Cl}^- \\ 2,8,7 & & & 2,8,8 \end{array}$																
3d	<table><tr><td>low</td><td>no</td></tr><tr><td>high</td><td>no</td></tr></table>	low	no	high	no	Chloromethane CH ₃ Cl is covalent molecular because it contains non-metals in the compound and is a gas at room temperature indicating a low melting point. Covalent substances do not conduct in any state. Sodium chloride is ionic because it is a compound of a metals and non-metal. Ionic compounds are all have high melting points and do no conduct in the solid state.												
low	no																	
high	no																	
4a(i)	waste gases hot air impurities iron	Problem Solving: Completing diagram labels from passage																
4a(ii)	Iron would turn into solid below 1538°C	The iron needs to be molten for it to be removed easily from the blast furnace. If the iron metal were to freeze into a solid then it would have to be scraped out of the furnace.																
4b	Fe ²⁺ → Fe ³⁺ + e ⁻	The reduction step Fe ³⁺ + e ⁻ → Fe ²⁺ is in the data booklet. The reversal of the equation means the equation becomes oxidation																
5a(i)	14 days	phosphorus-32 content = 100%, Time= 0days phosphorus-32 content = 50%, Time= 14days } half-life = 14 days																

5a(ii)	42days	<table><tr><td>Mass (g)</td><td>No of half lives</td></tr><tr><td>20</td><td>0</td></tr><tr><td>10</td><td>1</td></tr><tr><td>5</td><td>2</td></tr><tr><td>2.5</td><td>3</td></tr></table>	Mass (g)	No of half lives	20	0	10	1	5	2	2.5	3	1 half life = 14 days 3 half lives = 42 days																					
Mass (g)	No of half lives																																	
20	0																																	
10	1																																	
5	2																																	
2.5	3																																	
5b	Beta β	$^{32}_{15}\text{P} \rightarrow ^{32}_{16}\text{S} + ^0_{-1}\text{e}$																																
6	Open Question	3 mark answer	2 mark answer	1 mark answer																														
		Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																														
7a	Carboxyl group	<table><tr><td>$-\text{O}-\text{H}$</td><td>$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>	$-\text{O}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$	hydroxyl group	carboxyl group																												
$-\text{O}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$																																	
hydroxyl group	carboxyl group																																	
7b(i)																																		
7b(ii)	Answer to include:	Propanoic acid has three carbons and butanoic acid has four carbons. As number of carbons increases, melting point increases.																																
8a(i)	Glowed very brightly	Magnesium is more reactive than zinc so would react faster and brighter.																																
8a(ii)	Faster reaction	Powders react faster than lumps as powder has lower particle size																																
8b	Magnesium	MgO contains Mg^{2+} ions $\therefore \text{Mg} \longrightarrow \text{Mg}^{2+} + 2\text{e}^{-}$ (oxidation)																																
9a	Same chemical properties Same general formula	Homologous series (e.g. alkanes, alkenes, cycloalkanes) have																																
9b	isomers	same molecular formula (formula)	different structural formula (structure)																															
9c(i)	One answer from:	<table><tr><td>The longer the carbon chain the longer the time to get through column</td><td colspan="2">Branched chain isomers take less time to travel through column compared to their straight chain isomers</td></tr></table>			The longer the carbon chain the longer the time to get through column	Branched chain isomers take less time to travel through column compared to their straight chain isomers																												
The longer the carbon chain the longer the time to get through column	Branched chain isomers take less time to travel through column compared to their straight chain isomers																																	
9c(ii)	Line between 1 st and 2 nd peak	Propane would take longer to pass through the column than ethane (1 st peak) as it has a longer chain length. Propane would pass through the column quicker than the 2 nd set of peaks (isomers of butane) as it has a shorter chain length.																																
10a(i)	<table><tr><td>aluminium</td><td>copper</td></tr><tr><td>aluminium</td><td>copper</td></tr><tr><td>sulphate</td><td>sulphate</td></tr></table>	aluminium	copper	aluminium	copper	sulphate	sulphate	In an electrochemical cell, a metal is placed in a solution of that metal. Aluminium must be the metal on the left of the cell and copper the metal the right as electrons flow from the left to the right. This is because aluminium is higher up the electrochemical series than copper.																										
aluminium	copper																																	
aluminium	copper																																	
sulphate	sulphate																																	
10a(ii)	$3\text{Cu}^{2+} + 2\text{Al} \rightarrow 3\text{Cu} + 2\text{Al}^{3+}$	<table><tr><td>①</td><td>Al</td><td>$\rightarrow$</td><td>Al^{3+}</td><td>$+$</td><td>3e^{-}</td></tr><tr><td>②</td><td>Cu^{2+}</td><td>$+$</td><td>2e^{-}</td><td>$\rightarrow$</td><td>Cu</td></tr><tr><td>①x2</td><td>2Al</td><td>$\rightarrow$</td><td>2Al^{3+}</td><td>$+$</td><td>6e^{-}</td></tr><tr><td>②x3</td><td>3Cu^{2+}</td><td>$+$</td><td>6e^{-}</td><td>$\rightarrow$</td><td>3Cu</td></tr><tr><td>①'+②'</td><td>2Al</td><td>$+$</td><td>3Cu^{2+}</td><td>$\rightarrow$</td><td>$2\text{Al}^{3+} + 3\text{Cu}$</td></tr></table>			①	Al	$\rightarrow$	Al^{3+}	$+$	3e^{-}	②	Cu^{2+}	$+$	2e^{-}	$\rightarrow$	Cu	①x2	2Al	$\rightarrow$	2Al^{3+}	$+$	6e^{-}	②x3	3Cu^{2+}	$+$	6e^{-}	$\rightarrow$	3Cu	①'+②'	2Al	$+$	3Cu^{2+}	$\rightarrow$	$2\text{Al}^{3+} + 3\text{Cu}$
①	Al	$\rightarrow$	Al^{3+}	$+$	3e^{-}																													
②	Cu^{2+}	$+$	2e^{-}	$\rightarrow$	Cu																													
①x2	2Al	$\rightarrow$	2Al^{3+}	$+$	6e^{-}																													
②x3	3Cu^{2+}	$+$	6e^{-}	$\rightarrow$	3Cu																													
①'+②'	2Al	$+$	3Cu^{2+}	$\rightarrow$	$2\text{Al}^{3+} + 3\text{Cu}$																													
10b	15.8	$\text{gfm Al}_2(\text{SO}_4)_3 = (2 \times 27) + (3 \times 32) + (12 \times 16) = 54 + 96 + 192 = 342$ $\% \text{Al} = \frac{\text{total mass of Al}}{\text{gfm Al}_2(\text{SO}_4)_3} \times 100 = \frac{(2 \times 27)}{342} \times 100 = \frac{54}{342} \times 100 = 15.8\%$																																

11a	Answer to include:	Non-metal oxides like SO ₂ dissolve in water to form acids. As acid is formed the pH number will decrease below 7												
11b	As temp increases solubility decreases	Problem Solving: Drawing conclusion from a line graph												
12a	C=C double bond or -OH group	C=C double bonds are the functional groups found in alkenes Hydroxyl -OH groups are the functional group found in alcohols												
12b	Esters	Geraniol is an alcohol due to its -OH hydroxyl group. Alcohols react with carboxylic acid to form esters e.g. geraniol + propanoic acid → geranyl propanoate + water alcohol + carboxylic acid → ester + water												
12c	21.0g	1mol geraniol = (10×12)+(18×1)+(1×16) = 120+18+16 = 154g no. of mol = $\frac{\text{mass}}{\text{gfm}}$ = $\frac{15.4}{154}$ = 0.1mol geraniol + propanoic acid → geranyl propanoate + water C ₁₀ H ₁₈ O + C ₃ H ₆ O ₂ → C ₁₃ H ₂₂ O ₂ + H ₂ O 1mol 1mol 0.1 mol 0.1mol 1mol geranyl propanoate = (13×12)+(22×1)+(2×16) = 156+22+32 = 210g mass = no. of mol × gfm = 0.1 × 210 = 21.0g												
13a	C _n H _{2n-2}	<table><tr><td>Alkyne</td><td>Propyne</td><td>But-1-yne</td><td>Pent-1-yne</td></tr><tr><td>Molecular Formula</td><td>C₃H₄</td><td>C₄H₆</td><td>C₅H₈</td></tr><tr><td>General Formula</td><td>If n=3 2n-2 = (2×3)-2 = 4 ∴ C_nH_{2n-2}</td><td>If n=4 2n+2 = (2×4)-2 = 6 ∴ C_nH_{2n-2}</td><td>If n=5 2n+2 = (2×5)-2 = 8 ∴ C_nH_{2n-2}</td></tr></table>	Alkyne	Propyne	But-1-yne	Pent-1-yne	Molecular Formula	C ₃ H ₄	C ₄ H ₆	C ₅ H ₈	General Formula	If n=3 2n-2 = (2×3)-2 = 4 ∴ C _n H _{2n-2}	If n=4 2n+2 = (2×4)-2 = 6 ∴ C _n H _{2n-2}	If n=5 2n+2 = (2×5)-2 = 8 ∴ C _n H _{2n-2}
Alkyne	Propyne	But-1-yne	Pent-1-yne											
Molecular Formula	C ₃ H ₄	C ₄ H ₆	C ₅ H ₈											
General Formula	If n=3 2n-2 = (2×3)-2 = 4 ∴ C _n H _{2n-2}	If n=4 2n+2 = (2×4)-2 = 6 ∴ C _n H _{2n-2}	If n=5 2n+2 = (2×5)-2 = 8 ∴ C _n H _{2n-2}											
13b(i)		The repeating units can be joined together and when the brackets are removed the original polymer is drawn.												
13b(ii)	Addition	Addition polymerisation usually involves a C=C double bond opening up to leave a C-C single bond as the monomers join together to form the polymer. The C≡C triple bond in ethyne opens up to leave a C-C double bond as the monomers join together to form the polymer poly(ethyne)												
13c(i)		 + 2KOH → + 2KBr + 2H ₂ O												
13c(ii)	Bromines must be on adjacent carbons	The bromine atoms must be on carbons next door to each other if the triple bond is to form.												
14a(i)		<table><tr><td> Methanol</td><td> Ethanol</td><td> Propan-1-ol</td></tr><tr><td> Butan-1-ol</td><td> Pentan-1-ol</td><td> Hexan-1-ol</td></tr></table>	 Methanol	 Ethanol	 Propan-1-ol	 Butan-1-ol	 Pentan-1-ol	 Hexan-1-ol						
 Methanol	 Ethanol	 Propan-1-ol												
 Butan-1-ol	 Pentan-1-ol	 Hexan-1-ol												

14a(ii)	188kJ	Alcohol	Propan-1-ol	Butan-1-ol	Pentan-1-ol	Hexan-1-ol	Heptan-1-ol
		Energy Released (kJ)	158	170	179	185	-
		Difference		18kJ	9kJ	6kJ	(3kJ)
		Prediction	-	-	-	-	188kJ
14b	3.91	$E_h = cm\Delta T \therefore c = \frac{E_h}{m \times \Delta T} = \frac{13.3}{(0.1 \times 34)} = 3.91 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$					
15	Open Question	3 mark answer		2 mark answer		1 mark answer	
		Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.	

2017 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	1	7	1	91	0.1	0.0
2	C	4	4	90	2	0.0	0.0
3	A	67	19	12	2	0.2	0.0
4	D	13	11	17	60	0.1	0.0
5	A	58	28	8	5	0.2	0.0
6	B	11	72	12	5	0.2	0.0
7	C	2	3	86	9	0.1	0.0
8	B	13	70	9	7	0.5	0.0
9	A	56	20	8	16	0.1	0.0
10	A	85	12	2	1	0.0	0.0
11	D	1	11	4	83	0.1	0.0
12	C	12	4	50	34	0.1	0.0
13	C	9	3	81	8	0.1	0.0
14	B	5	92	2	1	0.0	0.0
15	D	7	10	7	77	0.1	0.0
16	C	6	7	79	8	0.2	0.1
17	A	70	4	9	16	0.1	0.0
18	B	3	95	1	2	0.0	0.0
19	A	64	6	17	12	0.4	0.0
20	C	5	29	40	26	0.1	0.1